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Tectonic rotation and diverse remanence carriers revealed by the paleomagnetism of Eo-Oligocene turbidites from two Ultrahelvetic units (Western Alps, Haute-Savoie, France)

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Abstract

Our paleomagnetic study of two Ultrahelvetic units from the Western Alps shows that: (1) one unit has experienced an important vertical-axis tectonic rotation, and (2) in turbidite deposits, the nature of the remanence carrier can differ between coarse-grained and fine-grained intervals.

Samples were collected at close intervals from weakly deformed turbidites in two different areas (Dérochoir and Marais). Characteristic remanent magnetization (ChRM) was measured with alternating field and thermal demagnetization, whereas clues for identifying the main remanence carrier were provided by coercivity spectra analysis.

In the Dérochoir area, the bulk of the section consists of fine-grained rocks. Magnetic intensity values are small, directions are scattered and no polarity reversals were detected. Remanence was probably acquired during late diagenesis or regional tectonic deformation. In coarser lithologies, depositional ChRM has probably been preserved by an early phase of cementation. In the Marais turbidites, remanent magnetization is mainly carried by magnetite. Although some samples show clustered directions, several polarity inversions have been observed, suggesting that remanence was acquired soon after sediment deposition. Magnetic directions values indicate a counterclockwise rotation of about 110° of the Marais sediments with respect to the stable European craton. This movement, which partly affects the underlying Subalpine chains, is probably related to sinistral strike–slip motion along a major regional accident during the Late Oligocene or Miocene. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: Paleogene; paleomagnetism; Prealps; rotation; turbidite; Western Alps

1. Introduction

The primary aim of this work was to improve our understanding of the stratigraphy and sedimentology of upper Eocene–lower Oligocene turbidite beds exposed in two Ultrahelvetic units from Haute-Savoie (Dérochoir and Marais areas), using a method (magnetostratigraphy) that had never been tested previously in these regions. This goal was not fully attained, but here we provide valuable information about the preservation of remanent magnetism in deformed turbiditic rocks, and further refine the tectonic history in one of the studied regions. Previous studies in surrounding structural units (Prealpine turbidites and Helvetic limestones) concluded that natural remanent magnetization (NRM) was acquired viscously

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(Channell et al., 1980; Kligfield and Channell, 1981). In contrast, our work reveals that the Marais turbidites have retained their depositional remanence, whereas the Dérochoir sediments were magnetized during late diagenesis or regional tectonic deformation. In addition, paleomagnetic direction analyses indicate that the Marais area has experienced an important vertical-axis tectonic rotation, confirming evidence from microtectonic data collected by R.W.H. Butler (1992) from underlying structural units.

2. Geological setting

In Haute-Savoie, so-called Ultrahelvetic units occur above the Morcles nappe (Helvetic realm), represented by the northern Subalpine chains (Haut-Giffre, Platé and Bornes massifs; Epard, 1990), and below the main body of the Prealps (Médianes and Breccia nappes) (Fig. 1). These kilometer-sized rock bodies consist of cover sediments derived from the former European margin of the Tethys (South-Helvetic and Valais domains; e.g. Rosset et al., 1976; Kindler, 1988) that were translated at least 50 km with respect to their inferred original basement, probably during the Oligocene (Ramsay, 1989). Following their emplacement over the Helvetic nappes, Ultrahelvetic units have been passively deformed by the tectonic events affecting the latter. The relative importance of tectonic slicing versus gravity gliding in the emplacement of the Ultrahelvetic units from Haute-Savoie is still controversial (e.g. Badoux, 1988; Jeanbourquin, 1992, 1994; Jeanbourquin et al., 1992; Piguet et al., 1998).

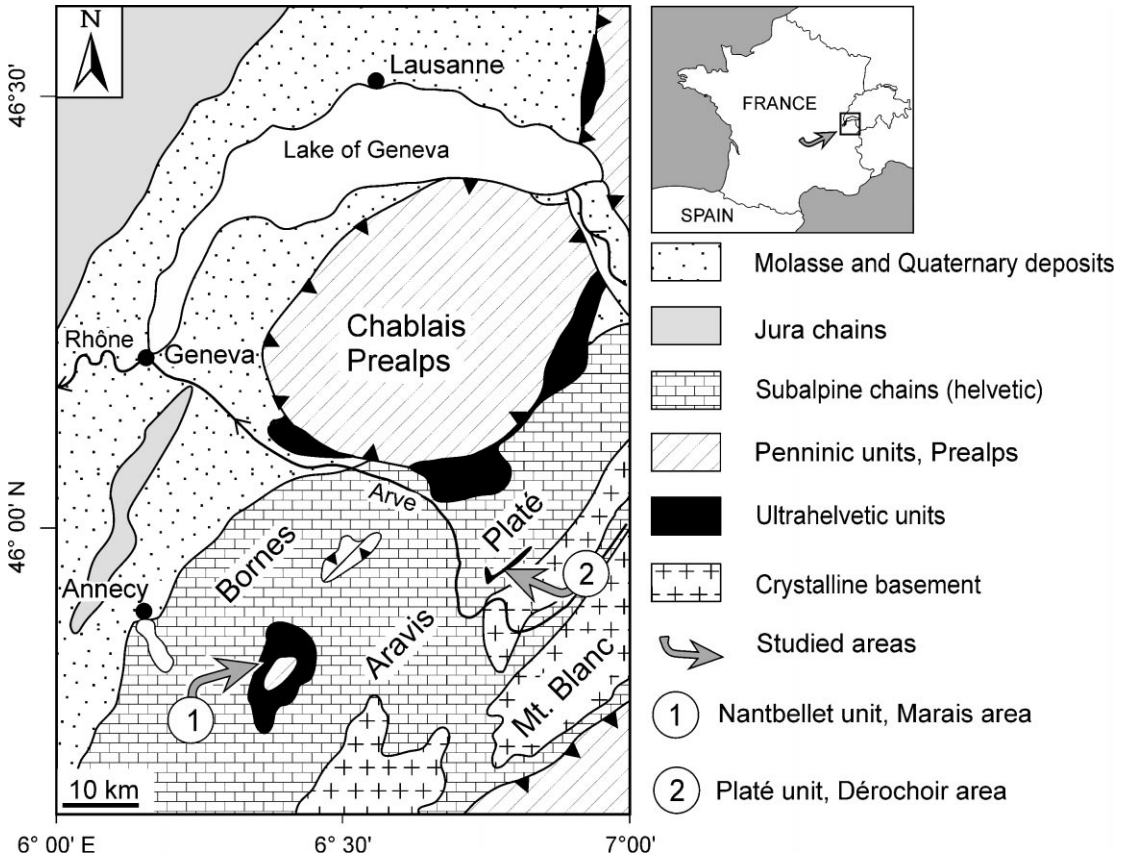


Fig. 1. Geological and tectonic setting of the studied areas.

Considering the possible disturbances of magnetic signals in deformed rocks, we deliberately chose to study two units where penetrative deformation is relatively weak: the Dérochoir unit, in the Platé massif and the Nantbellet unit, in the Sulens klippe (Fig. 1), where pilot samples had shown a stable remanence and the existence of magnetic reversals.

2.1. Dérochoir area

The Dérochoir unit is a kilometer-sized slide (Piguet et al., 1998), mainly consisting of shales, marls and siltstones, deposited by low-density turbidity and bottom currents. These fine lithologies further include channelized sandstones, debris-flow deposits, and isolated olistoliths in their upper part (Kindler, 1990; Piguet et al., 1998). Planktonic foraminiferal assemblages were identified in thin section and suggest a middle Eocene (P13–P14) to early Oligocene age for these sediments (Wernli et al., 1997), which unconformably overlie the lower Oligocene Taveyannaz Sandstones, the uppermost stratigraphic unit of the underlying Morcles nappe.

The Dérochoir unit was probably emplaced over the Taveyannaz Sandstones during the early Oligocene, just after the sedimentation of the latter, and was then passively affected by the folding of the Morcles nappe. A slaty cleavage was generated during this deformation phase (probably dating from the late Oligocene; Epard, 1990). By contrast, late tectonic events, including fracturing and large-scale folding related to the uplift of Mont Blanc massif, did not particularly affect the Dérochoir area.

2.2. Marais area

In the Nantbellet unit, a kilometer-sized thrust slice, we focused on the 180-m-thick well-bedded succession of turbidites and hemipelagites exposed in the Marais creek section. This shaly and sandy formation overlies pelagic marly limestones of middle Eocene age (Kindler et al., 1995; Ujetz, 1996). Planktonic foraminifer and nannofossil assemblages indicate a late Eocene age for the lower part of the Marais turbidites (Fig. 2; R. Wernli and K. von Salis, pers. commun.).

The tectonic units comprising the Sulens klippe have undergone a complex deformational history, including three main compressive phases of folding, and two extensional phases expressed by normal faults. The Nantbellet unit shows small-scale (i.e. wavelength of about 1 km) folds with variable axis directions. These structures occur above a major thrust plane (Fig. 3), and are thus not related to those of the underlying Helvetic zone. They have been interpreted as transported folds (Doudoux et al., 1992). In the sampling area (SW part of the Nantbellet unit), small-scale fold axes trend towards N355° and do not show any plunge.

In this region, the Subalpine chains show two generations of folding (Gidon, 1996) with (1) medium-scale (i.e. wavelength of about 3 km) folds, whose axes have been strongly curved (Fig. 3), and (2) large-scale (i.e. wavelength of more than 10 km) folds trending towards N30° and represented in the Marais area by the Serraval syncline (Fig. 3). The curvature of the medium-scale folds has formed a spoon-shape depression, which comprises the Sulens klippe. The sampling site is thus only marginally affected by these folds. In addition, it is located very close to the axial trace of Serraval syncline and is therefore not folded by it. However, it is affected by the plunge of this fold which dips about 10° towards the NNE (Fig. 3).

The succession of tectonic events leading to the aforementioned structures (detailed in Gidon, 1996) began with an extensional phase of Paleogene age which generated normal faults in the Subalpine chains. The latter were then overthrust by the already folded Ultrahelvetic (Nantbellet) and Prealpine units. This thrusting induced shear cleavage in the Subalpine chains and probably occurred during the late Oligocene. It was immediately followed by medium-scale folding of the area (although an early folding phase probably took place during the previous tectonic events). The late Miocene is marked by a second extensional phase producing normal and dextral strike–slip faults (Chaplet, 1989). Lastly, the uplift of the Belledonne massif induced the large-scale Serraval syncline.

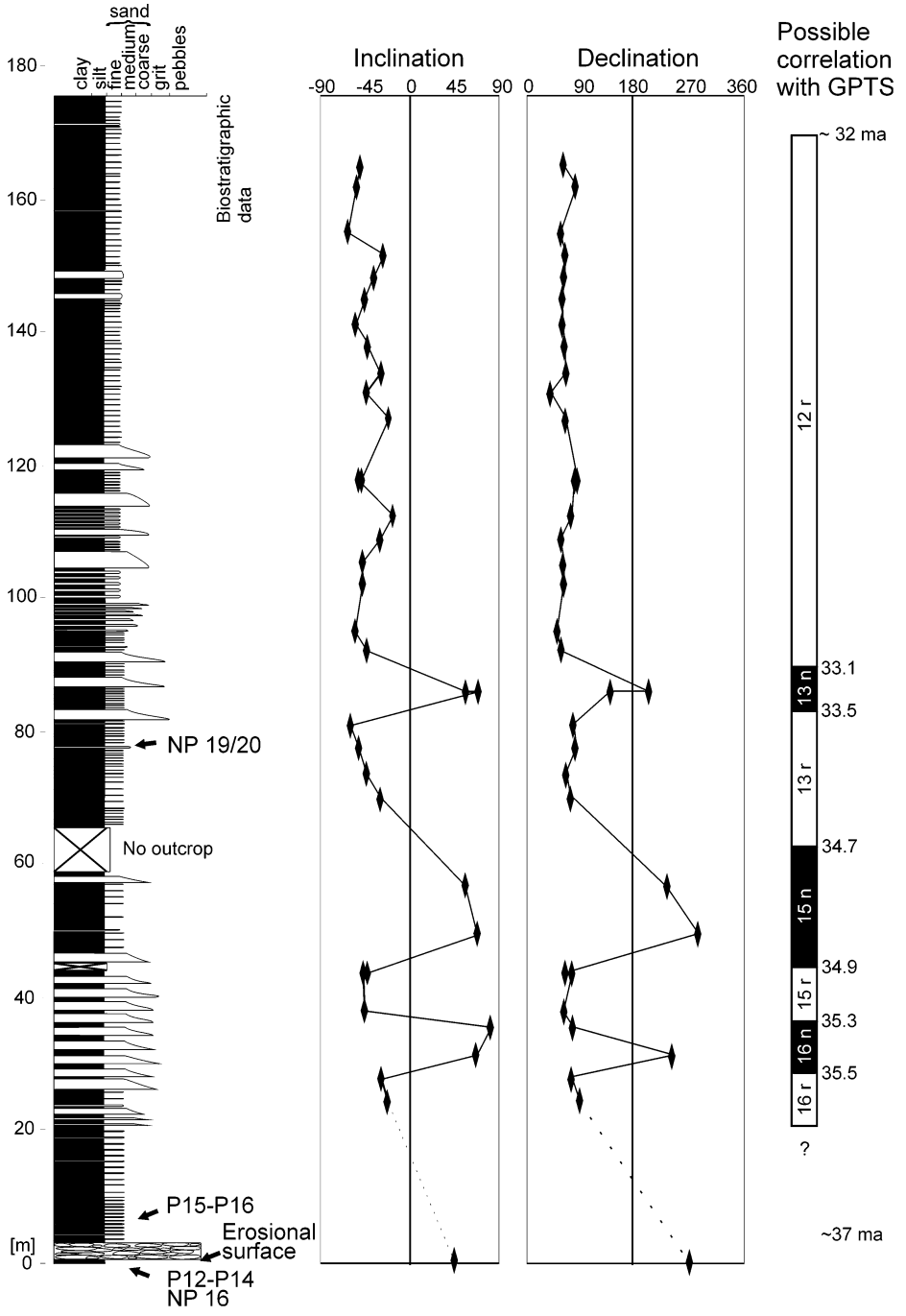


Fig. 2. Stratigraphy log and magnetostratigraphy of the Marais section and possible correlation with global polarity time scale (GPTS, Berggren et al., 1995).

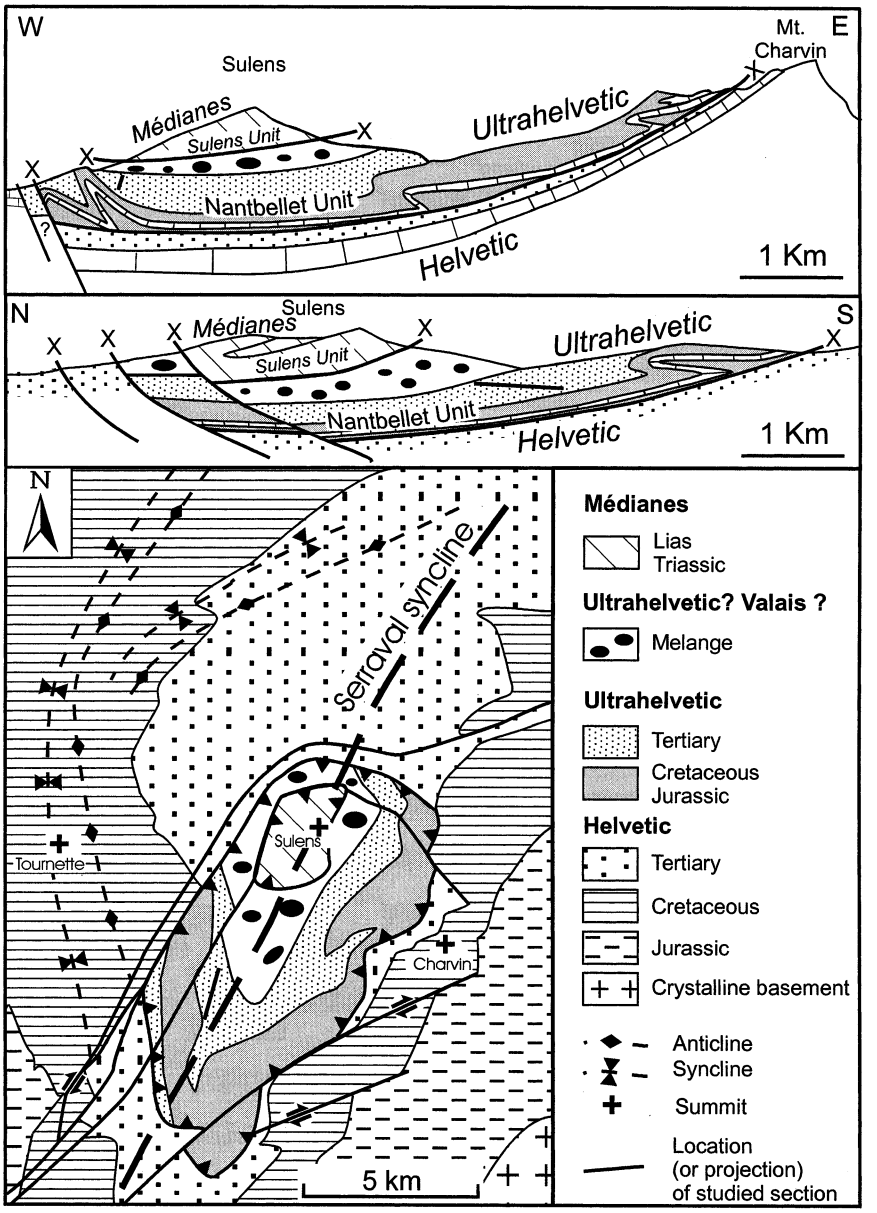
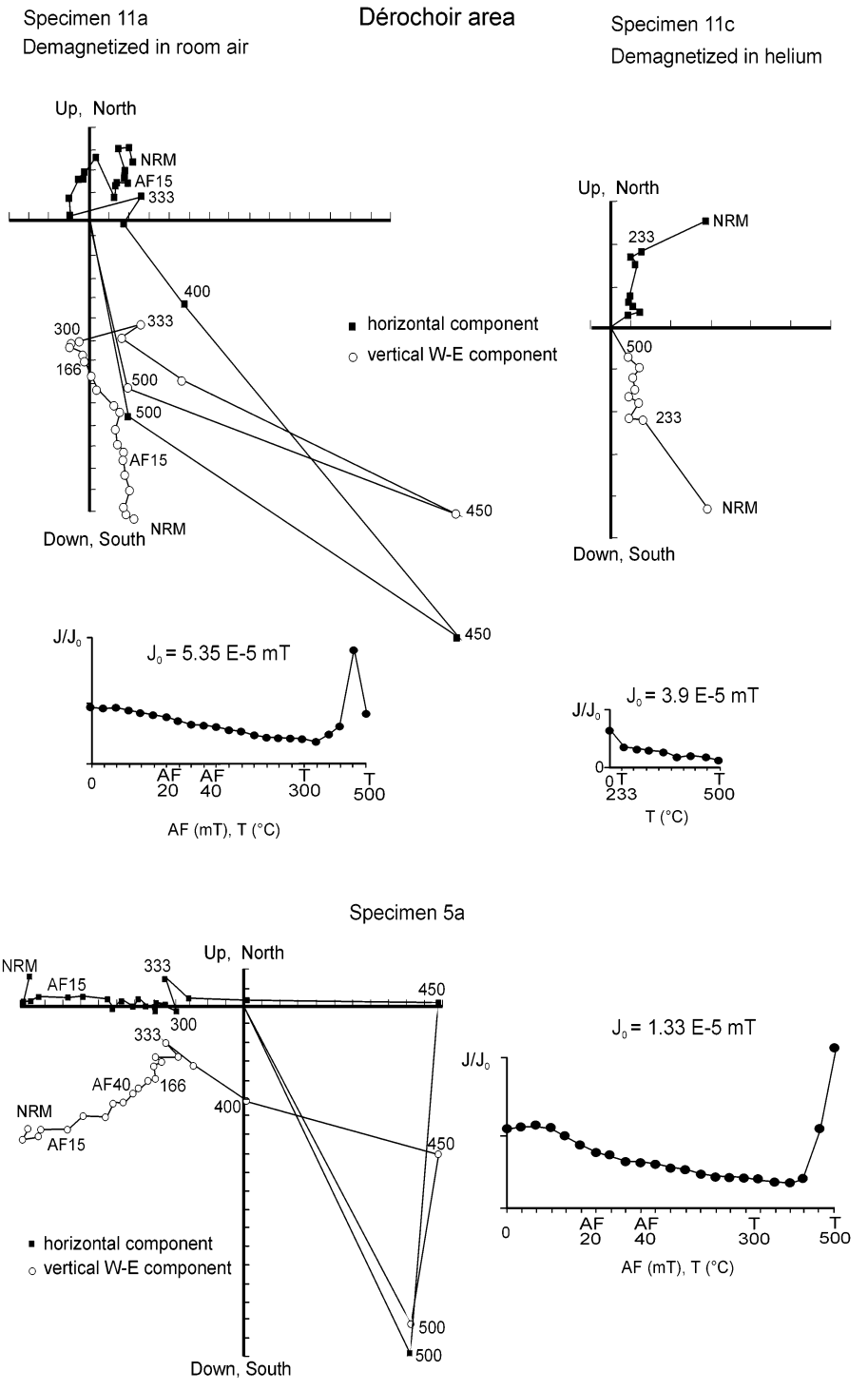


Fig. 3. Geological map and cross-section of the Marais area (after Doudoux et al., 1992; Gidon, 1996).

3. Methods

3.1. Sampling

The samples were collected as oriented blocks due to their fissile nature, cut into slabs of 2.5 cm^3 and coated with sodium silicate to avoid breakage during analysis. Special care was taken to collect very fine-grained and weakly deformed samples. In the Marais section, samples were collected along the continuous outcrop of the Marais creek. 60 sampling sites were selected about every 3 m in which at least two samples were gathered from the pelitic (Te) interval of Bouma sequences.


Dérochoir area
Specimen 11a
Demagnetized in room air
Specimen 11c
Demagnetized in helium
Up, North
NRM
AF15
333
400
300
333
166
500
500
AF15
NRM
450
450
Down, South
horizontal component
vertical W-E component
J/J0
J0 = 5.35 E-5 mT
0
AF 20
AF 40
T 300
T 500
AF (mT), T (°C)
Up, North
NRM
233
500
233
NRM
Down, South
J0 = 3.9 E-5 mT
J/J0
0
0 T 233
T 500
T (°C)
Specimen 5a
Up, North
NRM
AF15
333
300
450
333
AF40
166
400
NRM
AF15
450
horizontal component
vertical W-E component
500
500
Down, South
J0 = 1.33 E-5 mT
J/J0
0
AF 20
AF 40
T 300
T 500
AF (mT), T (°C)

30 sampling sites from the Dérochoir section were selected every meter and were also sampled twice in fine lithologies. Although strongest NRM intensities are usually measured on fine-grained sediments, Alvarez and Lowrie (1984) have shown that coarse turbiditic beds may record remanence directions that are indistinguishable from those of samples collected from overlying and underlying pelagic intervals. Therefore, we also sampled three pilot samples in fine-grained sandstone beds found in the upper part of the Dérochoir section because they are less deformed than the finer lithologies.

3.2. Coercivity spectra analysis

Cisowski (1981) used the intersection of isothermal remanent magnetization (IRM) against alternating field (AF) demagnetization to represent a median coercivity of a single-phase mineralogy. We use the coercivity spectral analysis to provide some indication of whether a high-coercivity phase exists in our samples. We realize it is difficult to estimate magnetic grain size from these data. Our coercivity results are tentatively compared with those of the magnetite standard for varying grain size ranges, and hematite standard and their mixture (Chang et al., 1987). Coercivity spectra analysis was used to quickly distinguish high-coercivity phases (like hematite and goethite) from low-coercivity phases (like magnetite and maghemite).

3.3. Demagnetization

NRM measurements were carried out using a cryogenic magnetometer at the University of Miami, Florida. Following NRM measurements, all samples were demagnetized with an alternating field from 5 to 40 mT with a 5 mT increment. Subsequently, they were heated from 100 to 500°C with a 33°C increment. Six samples showing signs of oxidation and a new ferrimagnetic phase during this process were submitted to a second simple thermal demagnetization in a helium atmosphere to prevent oxidation. We chose to use both AF and thermal demagnetization due to varying intensity of sample remanence. This combination allowed the identification of soft viscous components (using AF) as well as the more stable components. Magnetic susceptibility measurements were carried out on individual samples using a Bartington magnetic susceptibility bridge after each demagnetization step.

Principal component analysis (Kirschvink, 1980) was used to calculate the direction and associated maximum angular deviation (MAD) for each magnetization isolated in a specimen over a range of demagnetization steps. For the Marais section, magnetizations having MAD values greater than 10° were rejected. For the Dérochoir section, samples having MAD values greater than 15° were rejected. Greater MAD values were accepted for the Dérochoir section because of the weakness of magnetic intensities at this site which has resulted in more scattered directions. An area mean was calculated for all retained specimens in which a characteristic remanent magnetization (ChRM) could be defined using Fisher's (1953) statistics.

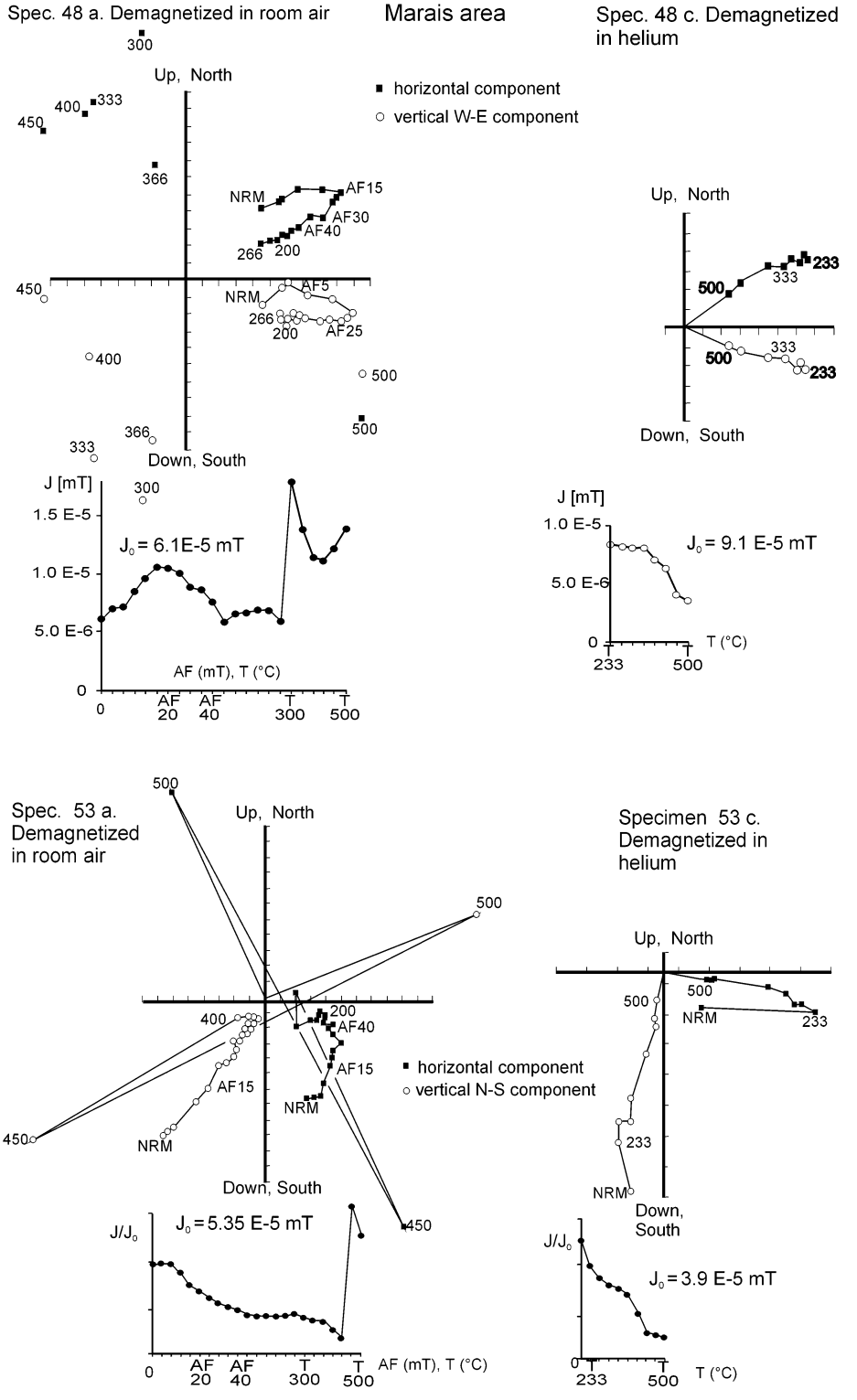
4. Magnetic properties

4.1. Demagnetization

4.1.1. Dérochoir

The majority of NRM intensities measured on fine-grained samples collected from the Dérochoir section were between 5×10^{-4} and 3×10^{-3} A/m. A few specimens were weakly magnetized and the magnetization plummeted to near the noise level of the cryogenic magnetometer upon demagnetiz-

Fig. 4. Results of AF and thermal demagnetization for representative samples from the Dérochoir area (in situ coordinate). Specimen 11a defines a vector that does not decay to the origin and which is affected by fluctuating directions at high temperature. Specimen 11c from the same sample has been demagnetized in helium and shows directions that do not correspond to those of specimen 11a. Specimen 5a shows a vector that decays linearly to the origin of the vector component diagram. Directions observed are considered as ChRM although high temperatures are affected by fluctuating direction caused by partial oxidation.


Spec. 48 a. Demagnetized in room air
Marais area
Spec. 48 c. Demagnetized in helium
Up, North
horizontal component
vertical W-E component
300
400
333
450
366
NRM
AF15
AF30
AF40
266
200
AF5
AF25
500
Down, South
J [mT]
1.5 E-5
1.0 E-5
5.0 E-6
0
J_0 = 6.1E-5 mT
AF (mT), T (°C)
AF 20
AF 40
T 300
T 500
J_0 = 9.1 E-5 mT
T (°C)
233
Spec. 53 a. Demagnetized in room air
Specimen 53 c. Demagnetized in helium
vertical N-S component
J/J_0
J_0 = 5.35 E-5 mT
J_0 = 3.9 E-5 mT

ing up to an AF field of 30–40 mT. These specimens usually yielded inconsistent directional data and were rejected. In about half of the samples from this area, NRM is composed of two or more components (Fig. 4, specimen 11a). After removal of an initial low-coercivity component in fields up to 20 mT (probably a viscous remanent magnetization showing a direction very close to that of the present-day field), demagnetization of the three component specimens defines a vector that generally does not decay to the origin of a vector component diagram until about 300°C. At these temperatures, a new direction begins to appear, but the production of new minerals at about 400°C results in a drastic increase in magnetic susceptibilities and intensities, accompanied by highly variable magnetic directions. This was probably related to the presence of sulfides within samples, which oxidize to hematite at temperatures higher than 400°C (R.F. Butler, 1992), or perhaps to the alteration of clay minerals. Specimens demagnetized in helium (Fig. 4, specimen 11c) show that a third high-temperature component is hidden by this oxidation effect. Unfortunately, fluctuating magnetic directions caused by oxidation do not permit us to confidently identify its magnetic direction.

Demagnetization of other specimens shows a vector that decays linearly to the origin until 400°C (Fig. 4, specimen 5a) and then scattered values caused by alteration at high temperatures. In this case, the stable directions observed below 400°C are considered as the ChRM.

The fine-grained sandstone samples have NRM intensities around 3×10^{-3} A/m which are higher than those observed in finer lithology. Despite scattered values at high temperatures, these beds mainly have a single magnetization component remaining after removal of the initial low-coercivity component.

4.1.2. Marais

Representative vector diagrams (Fig. 5) usually indicate the presence of a single magnetization component remaining after removal of initially low-coercivity components in fields of as much as 30–40 mT (but sometimes after heating at 166°C). At around 400°C, the production of new minerals during thermal demagnetization (in air) resulted in a drastic increase of magnetic susceptibilities and intensities, accompanied by fluctuating magnetic directions. As with the Dérochoir section, this increase is probably related to the presence of abundant sulfides or clay minerals within the samples. To assess whether a secondary component was hidden by these oxidation reactions at high temperature, six selected samples were thermally reprocessed (233–500°C) in a helium atmosphere. In three of these cases, demagnetization defines a vector that decays linearly with an orientation very similar to that of the first specimen of the same sample between 30 mT and 300°C (Fig. 5). In addition, susceptibilities did not increase during demagnetization. The three other samples do not show a stable demagnetization step and their susceptibilities increase again at a temperature of around 366°C. Thus, some alteration (clay minerals?) probably occurred in the latter despite processing in helium.

These additional samples show that the direction of the stable vector obtained after an AF demagnetization near 40 mT is the ChRM.

4.2. Magnetic mineralogy and coercivity spectra

The shape of the acquisition curves (sIRM) for samples collected from the Marais section (Fig. 6a, b) reveals the predominance of low-coercivity minerals. Median coercivity values of around 40 mT further suggest the presence of a mixture of coarse and ultrafine-grained magnetite. In contrast, the

Fig. 5. Results of AF and thermal demagnetization for representative samples from the Marais area (in situ coordinate). Specimens labeled "a" were demagnetized in room air and show a drastic increase in magnetic susceptibility and intensities between 300 and 400°C, accompanied by erratic magnetic directions. In contrast, specimens labeled "c" from the same samples were demagnetized in helium and show a decrease in intensity and an approximately unchanged susceptibility. Specimens demagnetized in helium further confirm the absence of an additional component hidden by sample oxidation at high temperature.

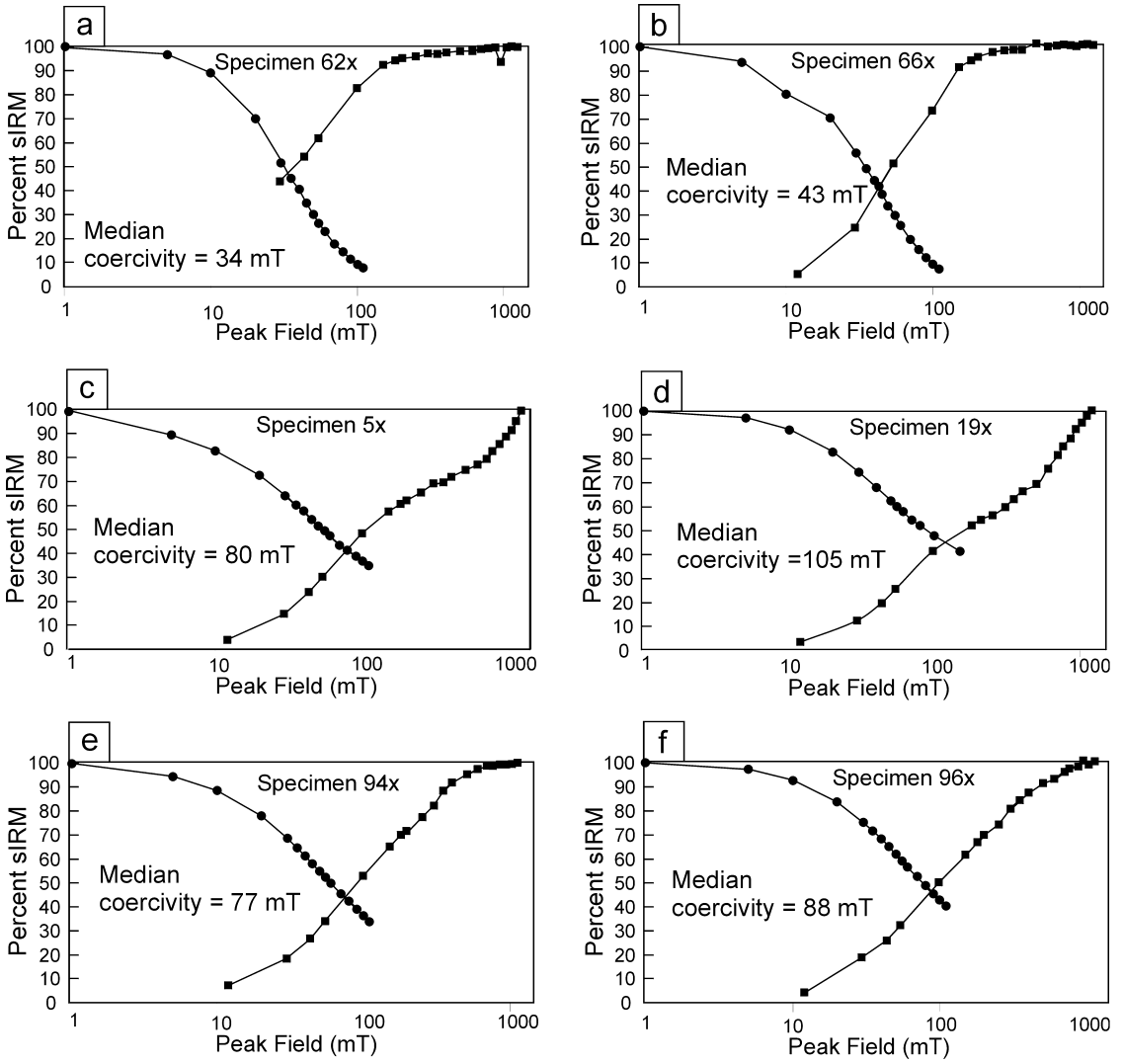


Fig. 6. Acquisition of IRM and AF demagnetization (coercivity spectra) of samples from Marais area (a, b), shales and silts from the Dérochoir area (c, d), fine-grained sandstones of the Dérochoir area (e, f).

shapes of the acquisition curves for the shales and silts from the Dérochoir section (Fig. 6c, d) show the presence of high-coercivity minerals. Median coercivity values of around 90 mT suggest that the principal magnetic mineral is hematite or goethite, probably mixed with some coarse-grained magnetite. Fine-grained sandstones from the same section show, however, a curve indicating the presence of minerals (Fig. 6e, f), with median coercivity values of around 80 mT. The main magnetic mineral occurring in these samples is probably coarse-grained magnetite, with some hematite driving coercivity values higher.

Based on the sIRM and AF/thermal demagnetization data, we propose that the Marais samples, and probably the fine-grained sandstone facies of the Dérochoir are dominated by magnetite. The fine-grained shales and siltstones of the Dérochoir

appear to have a greater proportion of a high-coercivity phase relative to the Marais samples and Dérochoir sandstones.

4.3. Remanent magnetization and paleomagnetic directions

4.3.1. Dérochoir

The ChRM directions of fine-grained specimens with only two components are scattered and no reverse polarities were encountered in this section (Fig. 7a, b and Table 1). Bedding plane and slaty cleavage orientations are shown in Fig. 7c and d and mean site directions in Table 2. ChRM directions yield a mean declination (D) of 312.3 and an inclination of 58.5 in tilt-corrected (stratigraphic) coordinates, with a confidence limit (α_{95}) = 11.4 and dispersion estimate (K) = 9.5. ChRM directions of these specimens before tilt correction (geographic) yield a mean declination of 294.7 and an inclination of 37.2 with $\alpha_{95} = 12.1$ and K = 8.4. Despite a slight decrease in dispersion after tectonic correction, a fold test could not be applied in this area because of the small variation in tilt orientation. Pilot specimens in fine-grained sandstones show normal ChRM directions after tilt correction (Fig. 7a, b).

4.3.2. Marais

ChRM directions and bedding plane orientations from the Marais section are shown in Fig. 8 and Table 3, whereas mean site directions are presented in Table 2. 11 out of 60 demagnetized samples have MAD values higher than 10° and

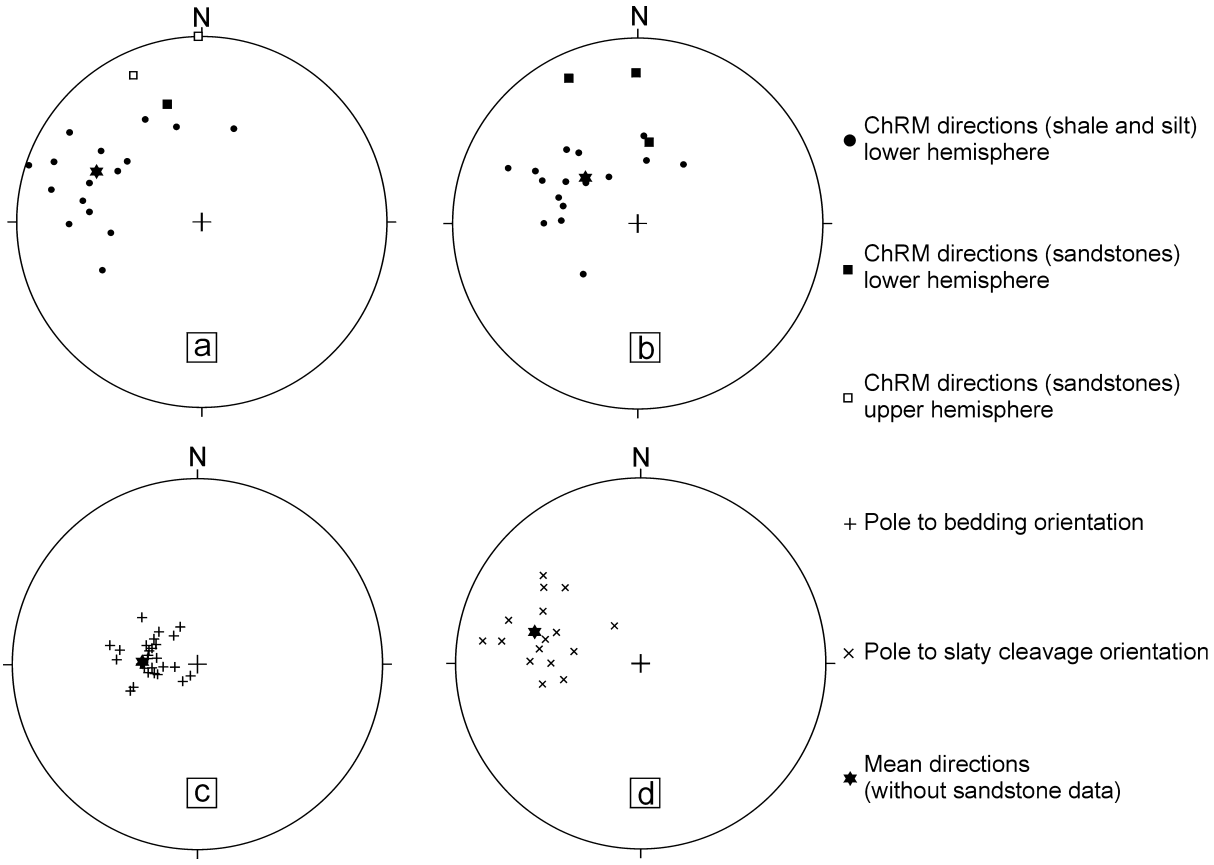


Fig. 7. Equal area projection of ChRM directions and bedding plane orientations for the Dérochoir area: (a) in situ, (b) tilt-corrected directions, (c) pole to bedding plane orientations, (d) pole to slaty cleavage orientations. Site statistics are given in Table 2.

Table 1
ChRM directions, Dérochoir section

Specimen	Thickness (m)[a]	G. Dec	G. Inc.	S. Dec	S. Inc.	Strike/Dip[b]
Shale and silt						
3a	2.5	301	46	310	60	019/21
4a	3.75	309	47	328	66	015/22
5a	5	275	39	283	56	345/18
6a	6.25	289	36	300	53	350/20
9a	10	345	46	8	62	040/22
10a	11.25	282	18	294	43	338/32
11c	12.5	19	46	38	57	065/18
13a	15	304	15	316	44	003/36
18a	20.5	331	37	4	51	010/35
19a	21.75	244	40	227	57	006/22
20a	23	288	2	293	25	340/30
21a	24.25	263	49	272	56	310/10
23b	26.75	305	34	320	49	355/22
24a	28	280	35	288	53	350/20
27a	31.75	292	15	297	38	000/25
28a	33	269	29	270	48	355/20
Fine sandstones						
92a	40	344	35	9	53	030/30
95a	42	335	−15	335	14	048/30
97a	44	359	−1	0	20	070/22

[a] With respect to the bottom of the section.
[b] Strike is recorded to the left of the downdip direction.

were therefore rejected (Kirschvink, 1980). 35 of the 49 remaining specimens comprise two groups of reversed direction in tilt-corrected coordinates (seven with positive inclination, 28 with negative inclination, Fig. 8b) and 14 other specimens are scattered. ChRM directions of the latter samples are probably dispersed because of the detrital orientation acquired by coarser magnetic grains during turbidite deposition. These 14 specimens were rejected as well (if these samples had not been rejected, data would be more scattered but mean site directions would not change significantly). Among the 35 retained specimens (Fig 8c, d), ChRM directions with positive inclination

Table 2
ChRM means and site statistics

Site		N/n	Dec.	Inc.	a^{95}	K	R
Dérochoir		16/30					
	In situ		294.7	37.2	12.1	8.4	14.2
	Tilt-corrected		312.3	58.5	11.4	9.5	14.4
Marais		35/60					
	Normal polarity ($n=7$)						
	In situ		265.3	18.3	21.2	6.3	6.0
	Tilt-corrected		247.8	70.8	14.1	14.1	6.6
	Reverse polarity ($n=28$)						
	In situ		58.4	7.5	6.7	15.4	26.3
	Tilt-corrected		62	−43.2	4.8	30	27.1

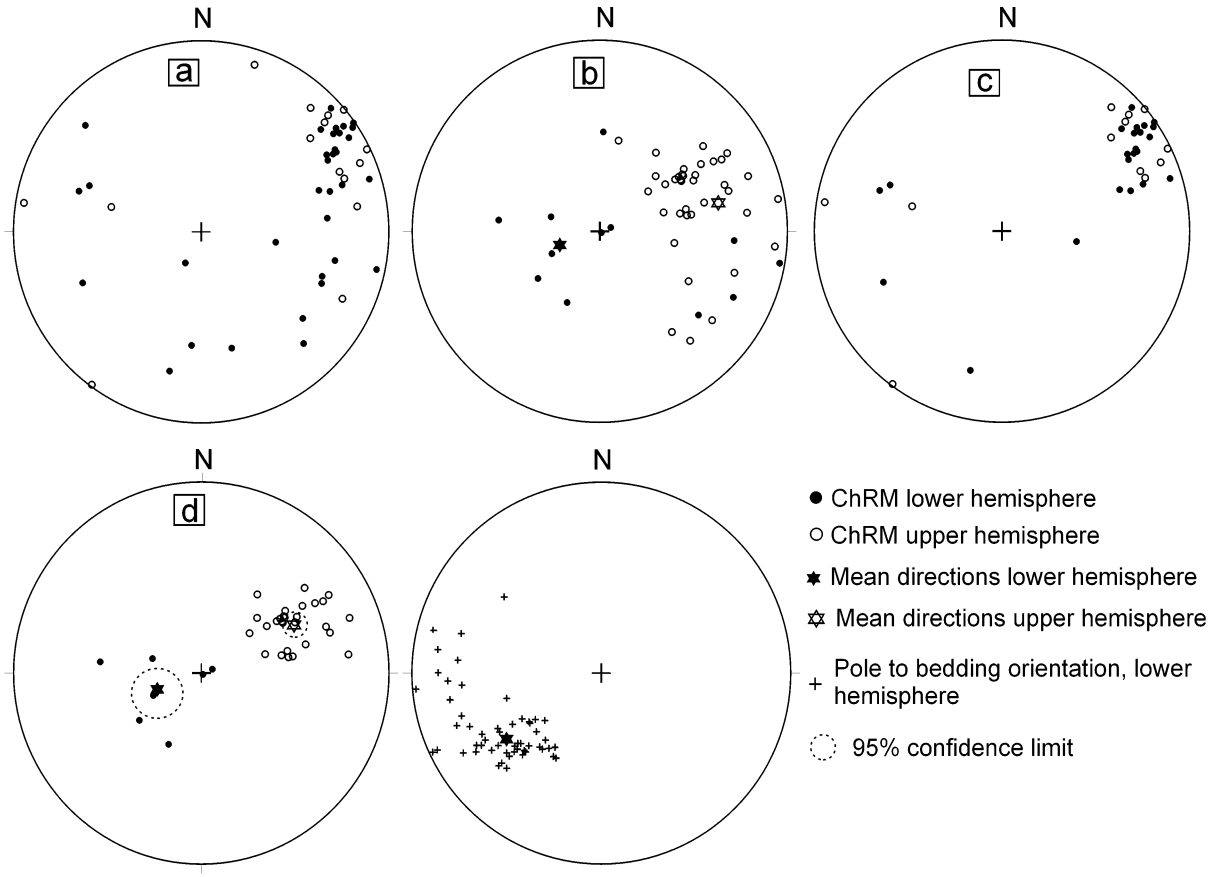


Fig. 8. Equal area projection of ChRM directions and bedding plane orientations for the Marais area: (a) all samples, in situ; (b) all samples, tilt-corrected directions; (c) in situ remaining samples; (d) tilt-corrected remaining samples; (e) pole to bedding plane orientations. Site statistics are given in Table 2.

(seven specimens) yield a mean declination (D) of 247.8 and an inclination of 70.8 in tilt-corrected (stratigraphic) coordinates, with $\alpha_{95}=14.1$ and $K=14.1$. Before tilt correction (geographic), ChRM directions of these specimens yield a mean declination of 265.3 and an inclination of 18.3 with $\alpha_{95}=21.2$ and $K=6.3$. ChRM directions with negative inclination (28 specimens out of 35) yield a mean declination (D) of 62.0 and an inclination of -43.2 in tilt-corrected (stratigraphic) coordinates, with $\alpha_{95}=4.8$ and $K=30.0$. ChRM directions of these specimens before tilt correction (geographic) yield a mean declination of 58.4 and an inclination of 7.5 with $\alpha_{95}=6.7$ and $K=15.4$.

5. Discussion and interpretation

5.1. Dérochoir

In the Dérochoir turbidites, ChRM orientations are scattered (normal polarity) and do not show any polarity reversals. This was expected in the fine sandstones due to the thinness of the studied horizon. In the thicker shaly and silty interval, whose age ranges from the middle Eocene to the early Oligocene (Wernli et al., 1997), ChRM directions were probably acquired after deposition during tectonic deformation or during late diagenesis (e.g. Reynolds et al., 1994). We favor the

Table 3
ChRM directions, Marais section

Specimen	Thickness (m)[a]	G. Dec	G. Inc.	S. Dec	S. Inc.	Strike/dip[b]
31b	1	285	−49	269	43	333/60
32a[c]	4	102	5	119	−45	338/64
34a[c]	9	110	32	107	−26	340/70
35a[c]	13	312	18	2	47	357/70
35b[c]	13	137	21	140	−26	357/70
37a[c]	18	130	30	128	−25	005/65
39a[c]	24	185	40	130	32	343/66
40a[c]	28	165	37	116	22	330/73
42a	35	98	57	83	−22	335/85
43b	36	79	58	73	−29	335/88
44a	40	292	36	245	67	047/45
45a	42	288	32	73	85	013/65
46a	45	54	14	57	−46	315/60
48a	46	72	22	75	−42	330/65
48c	46	60	18	59	−48	330/65
49a	50	279	−5	286	68	000/75
51a[c]	51	207	75	94	29	350/70
52a	55	247	32	233	56	355/88
53a[c]	59	102	28	95	−7	320/50
54a[c]	60	113	31	100	3	320/48
55a[c]	64	81	−17	99	−57	330/45
57a	69	71	34	70	−30	330/65
58a	70	61	24	62	−43	328/67
59a	78	73	7	80	−51	330/60
60a	79	64	−2	74	−61	325/60
61a	83	193	26	137	89	318/60
61b	83	216	−1	205	56	318/60
62a	87	53	11	54	−44	318/55
63a	90	47	6	46	−56	318/62
65a[c]	97	115	−18	144	−35	325/56
66a	100	54	7	56	−48	320/55
67a	101	55	1	58	−48	318/50
69a	105	50	18	51	−30	315/48
70b	108	60	20	70	−17	310/38
71a	109	67	−21	76	−51	320/32
71b	109	70	−20	80	−49	320/32
72a[c]	113	84	33	80	−4	330/40
72b	113	59	18	59	−22	330/40
74a	118	42	−14	36	−48	325/35
75a	119	58	8	59	−30	325/38
76a	123	56	3	60	−41	315/45
77a	127	49	−14	55	−55	310/42
78a	128	48	−10	56	−46	302/38
79a	132	55	11	57	−36	314/48
80a	133	59	23	60	−26	314/50
85a	147	50	−25	51	−63	318/38
86a[c]	151	18	−9	12	−50	298/42
89a	161	67	−9	78	−54	320/48
90a	165	50	−1	56	−50	308/42

[a] With respect to the bottom of the section.
[b] Strike is recorded to the left of the downdip section.
[c] Rejected samples.

former hypothesis because ChRM directions are approximately normal to the orientation of the observed slaty cleavage (Fig. 7d) and could correspond to the orientation of the axis of maximum shortening. In addition, microscopic observations reveal the occurrence of recrystallized tension gashes at a right angle to slaty cleavage planes, further suggesting that remanent magnetization is probably associated with regional deformation.

ChRM directions measured in pilot specimens from fine-grained sandstones differ from those obtained from shales and siltstones and have probably been acquired during (or shortly after) sedimentation by alignment of grains in the geomagnetic field. The contrast in the mineralogy of remanence carriers between shale/silt lithologies (mixed magnetite and high-coercivity phase) and fine-grained sandstones (magnetite) supports this interpretation.

ChRM directions of the Ultrahelvetic deposits from the Dérochoir area were thus acquired during both deposition (fine-grained sandstones) and deformation or alteration (silts and shales) of the sediments. The preservation of primary remanence in coarser lithologies is possibly related to an early phase of cementation (Piguet et al., 1998) which protected the sediment against shearing or alteration.

5.2. Marais

In this area, ChRM directions of tilt-corrected coordinates are grouped in both positive and negative inclinations. Using the fold test (McFadden and Jones, 1981), the hypothesis of a common mean direction for the two populations (18 sites in the NE limb and 10 sites in the SW limb; Fig. 3) can be rejected at the 99% confidence level for the in situ mean direction, but cannot be rejected for tilt-corrected directions (in situ observed value = 0.46, critical value at 99% confidence level = 0.19, tilt-corrected observed value = 0.032). Thus, the observed magnetic directions clearly pass the fold test, strongly suggesting that remanence has been acquired before structural tilting. However, tilt-corrected directions show a normal mean inclination of 71° which is too steep, and a reverse inclination of –43° which is shallower than expected (expected inclination would be around 62° for this area which was located near 43°N at the Eocene/Oligocene boundary; Stampfli et al., 1998). The angle between mean positive inclination and the antiparallel vector of mean negative inclination is about 29°. These data failed the reversal test (McFadden and McElhinny, 1990), either because of (1) a tectonic effect uncorrected by tilt correction (e.g. the weak axial plunge of the Serraval syncline), (2) an early post-depositional effect that deviated directional data, or (3) strong overprinting which was not removed during the demagnetization procedure. As shown by specimen cooling in helium, high-temperature overprint (if present) does not have a significant influence on the primary magnetic direction, but uncorrected tectonic or early post-depositional effects cannot be totally excluded.

Despite the failure of the reversal test, suggesting that mean magnetic direction should be viewed cautiously (with an error envelope of a few degrees), ChRM directions reveal several polarity inversions along the Marais section, with three normal and three reverse polarity zones (Fig. 3). ChRM directions are thought to have been acquired soon after sediment deposition. In addition, calculated paleomagnetic directions imply a counterclockwise, vertical-axis rotation of the studied area approaching 110° with respect to known Tertiary declination values for the stable European plate which are close to those of the present magnetic field in this region (Heller et al., 1989). Considering the regional geological context, this rotation could result from tectonic transport in thrust units or from strike–slip action. Fold axes trend towards N355° in the SW part of the Nantbellet unit, whereas they trend towards N30° in the eastern portion of this unit (Doudoux et al., 1992). We propose that the SW part of Nantbellet unit underwent a counterclockwise vertical-axis rotation of about 35° during its emplacement over the subalpine chains. Furthermore, on the basis of microtectonic evidence, R.W.H. Butler (1992) suggested that the SW end of the Bornes massif (Subalpine chains) has been rotated by about 60°, and he explained this rotation by sinistral strike–slip movements along the Veyrier fault, a major tectonic accident located to the west of the studied

area. Rotation values deduced from fold-axis analysis in the Nantbellet unit (this paper) and microtectonic criteria in the underlying Subalpine chains (R.W.H Butler, 1992) add up to about 95°, a value close to that obtained from paleomagnetic direction analysis in the former unit. Thus, the vertical-axis rotation of the Nantbellet unit was probably initiated during its emplacement over the Subalpine chains and completed during rotation of the latter. Alternatively, the Nantbellet unit and the Subalpine chains were simultaneously rotated, but the former experienced more rotation due to a higher, less constrained structural position. As suggested by R.W.H. Butler (1992), these rotational movements are probably related to a major sinistral strike–slip zone affecting this sector of the Western Alps.

6. Conclusion

This study, the first paleomagnetic investigation in the Ultrahelvetic nappes from Haute-Savoie indicates that, in one of the two studied units (Nantbellet, Marais Creek section), primary (post-depositional) remanent magnetization is still preserved despite a subsequent complex tectonic history. In the other case (Dérochoir), remanent magnetization was probably acquired during late diagenesis or regional tectonic deformation. Magnetic minerals occurring in the studied turbidites vary according to lithology as suggested by coercivity spectra analysis. In the Marais region, the dominant NRM is probably carried by magnetite. In the Dérochoir area, a mixture of a high-coercivity phase, and probably some coarse-grained magnetite, characterize the shale and siltstones sediments, whereas a higher proportion of coarse-grained magnetite is found in coarser lithologies. These preliminary observations suggest that, in turbidite deposits, the origin of remanent magnetization can differ with respect to grain size and, wherever fine-grained sediments show a weak penetrative deformation, coarser-grained beds seem to be a better sampling target to recover primary remanent magnetization.

Most of the ChRM directions from the Marais area were probably acquired soon after deposition and record seven polarity zones. Direction values imply that the Marais area underwent a significant rigid counterclockwise rotation of about 110° with respect to the stable European craton. This movement, which partly affects the underlying Subalpine chains, is probably related to a major strike–slip zone in this area of the Western Alps. Paleomagnetic direction analysis is thus a powerful tool for unraveling multi-stage tectonic events, and future studies in the Marais area should contribute to better constrain the time when the Sulens Unit (Médianes Prealps) overthrust the Nantbellet unit and to establish whether these two units were rotated together.

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References

Alvarez, W., Lowrie, W., 1984. Magnetic stratigraphy applied to synsedimentary slumps, turbidites, and basin analysis: the Scaglia Limestone at Furlo (Italy). Geol. Soc. Am. Bull. 95, 324–336.

Badoux, H., 1988. L'Ultrahelvétique inférieur en Suisse occidentale. Bull. Soc. Frib. Sci. Nat. 77, 1–2, 143–152.

Berggren, W.A., Kent, D.V., Swisher, C.C., Aubry, M.-P., 1995. A revised Cenozoic geochronology and chronostratigraphy. In: Berggren, W.A., Kent, D.V., Aubry, M.-P., Hardenbol, J. (Eds.), Geochronology, Time Scales and Global Stratigraphic Correlation. Soc. Econ. Paleontol. Mineral., Spec. Publ. 54, 129–212.

Butler, R.F., 1992. Paleomagnetism. Blackwell Science, Boston, 319 pp.

Butler, R.W.H., 1992. Thrusting patterns in the NW French subalpine chains. Ann. Tectonicae VI (2), 150–172.

Chang, S.B.R., Kirschvink, J.L., Stolz, J.F., 1987. Biogenic magnetite as a primary remanence carrier in limestone deposits. Phys. Earth Planet. Inter. 46, 289–303.

Channell, J.E.T., Freeman, R., Heller, F., 1980. Magnetic properties of Swiss Flysch. J. Geophys. 48, 60–66.

Chaplet, M., 1989. Etude géologique du massif subalpin des Bornes (Haute-Savoie). Thèse, Univ. Savoie, Chambéry, France, 218 pp. (unpublished).

Cisowski, S., 1981. Interactive vs. non-interacting single domain behavior in natural and synthetic samples. Phys. Earth. Planet. Inter. 26, 56–62.

Doudoux, B., Rosset, J., Barféty, J.-C., Pairis, J.-L., 1992. Carte géologique de France (1/50 000), feuille Annecy-Ugine (702). Orléans, BRGM ed.

Epard, J.-L., 1990. La nappe de Morcles au sud-ouest du Mont-Blanc. Thèse, Mém. Géol. Lausanne, no. 8, 165 pp.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. London A 217, 295–305.

Gidon, M., 1996. Vues nouvelles sur la structure des massifs des Bornes et des Bauges orientales. Géol. Alpine 72, 35–59.

Heller, F., Lowrie, W., Hirt, A.M., 1989. A review of palaeomagnetic and magnetic anisotropy results from the Alps. In: Coward, M.P., Dietrich, D., Park, R.G. (Eds.), Alpine Tectonics. Geol. Soc. London, Spec. Publ. 45, 399–420.

Jeanbourquin, P., 1992. Les mélanges suprahelvétiques dans le synclinal de Thônes (massifs subalpins, Haute-Savoie, France): une nouvelle hypothèse de travail. Arch. Sci., Genève 45 (2), 109–126.

Jeanbourquin, P., 1994. Early deformation of Ultrahelvetic mélanges in the Helvetic nappes (Western Swiss Alps). J. Struct. Geol. 16 (10), 1367–1383.

Jeanbourquin, P., Kindler, P., Dallagnolo, S., 1992. Les mélanges dans les Préalpes internes entre Arve et Rhône (Alpes occidentales franco-suisses). Eclogae Geol. Helv. 85 (1), 59–83.

Kindler, P., 1988. Géologie des wildflyschs entre Arve et Giffre (Haute-Savoie, France). Publ. Dépt. Géol. Paléont. Univ. Genève, no 6. 134 pp.

Kindler, P., 1990. Géologie du sommet de la Pointe-de-Platé (Domaine helvetique, Haute-Savoie, France), Calcaire paléocènes et faciès chaotiques. Eclogae Geol. Helv. 83 (1), 7–19.

Kindler, P., Ujetz, B., Charollais, J., Wernli, R., 1995. Submarine resedimentation of Cretaceous limestone during the Paleogene: the Formation gréso-glauconieuse from the Ultrahelvetic Prealps (Haute-Savoie, France). Bull. Soc. Géol. Fr. 166 (5), 507–515.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophys. J. R. Astron. Soc. 62, 699–718.

Kligfield, R., Channell, J.E.T., 1981. Widespread remagnetization of Helvetic limestones. J. Geophys. Res. 86, B3, 1888–1900.

Lowrie, W., Hirt, A.M., Kligfield. R., 1986. Effect of tectonic deformation on the remanent magnetization of rocks. Tectonics 5 (5), 713–722.

McFadden, P.L., Jones, D.L., 1981. The fold test in palaeomagnetism. Geophys. J. R. Astron. Soc. 67, 53–58.

McFadden, P.L., McElhinny, M.W., 1990. Classification of reversal test in palaeomagnetism. Geophys. J. Int. 103, 725–729.

Piguet, B., Morend, D., Kindler, P., Sartori, M., 1998. Le complexe turbiditique de Platé (massif de Platé, Haute-Savoie, France): un olistolite de flysch sudhelvetique surmontant les Grès de Taveyannaz. Eclogae Geol. Helv. 91 (2), 261–273.

Ramsay, J.G., 1989. Fold and fault geometry in the western Helvetic nappes of Switzerland and France and its implication for the evolution of the arc of the western Alps. In: Coward, M.P., Dietrich, D., Park, R.G. (Eds.), Alpine Tectonics. Geol. Soc. London, Spec. Publ. 45., 33–45.

Reynolds, R.L., Tuttle, M.L., Rice, C.A., Fishman, N.S., Karachewski, J.A., Sherman, D.M., 1994. Magnetization and geochemistry of greigite-bearing Cretaceous strata, north slope basin, Alaska. Am. J. Sci. 294, 485–528.

Rosset, J., Charollais, J., Toumarkine, M., Manivit, H., Chateauneuf, J.J., Schaub, H., 1976. Présentation des différentes unités du synclinal de Thônes (Haute-Savoie, France). Eclogae geol. Helv. 69 (2), 359–402.

Stampfli, G.M., Mosar, J., De Bono, A., Vavasis, I., 1998. http://www-sst.unil.ch/Plate_tecto/Alp_Tet.htm.

Ujetz, B., 1996. Micropaleontology of Paleogene deep water sediments, Haute-Savoie, France. Publ. Dépt. Géol. Paléont. Univ. Genève 22, 149 pp.

Wernli, R., Morend, D., Piguet, B., 1997. Les foraminifères planctoniques en section de l'Eocène et de l'Oligocène des Grès de Samoëns (Ultrahelvétique du massif de Platé, Haute-Savoie, France). Eclogae Geol. Helv. 90, 581–590.